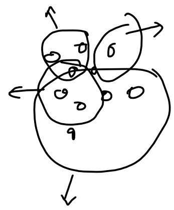
Kmeans → unsupervised

$K = 3, 5$



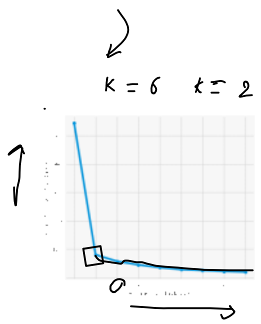
EM

$$J = \sum_{i=1}^{m} \sum_{k=1}^{K} r_{ik} \| x^{i} - \mu_{k} \|^{2}$$

Elbow → inertia

Silhouette analysis

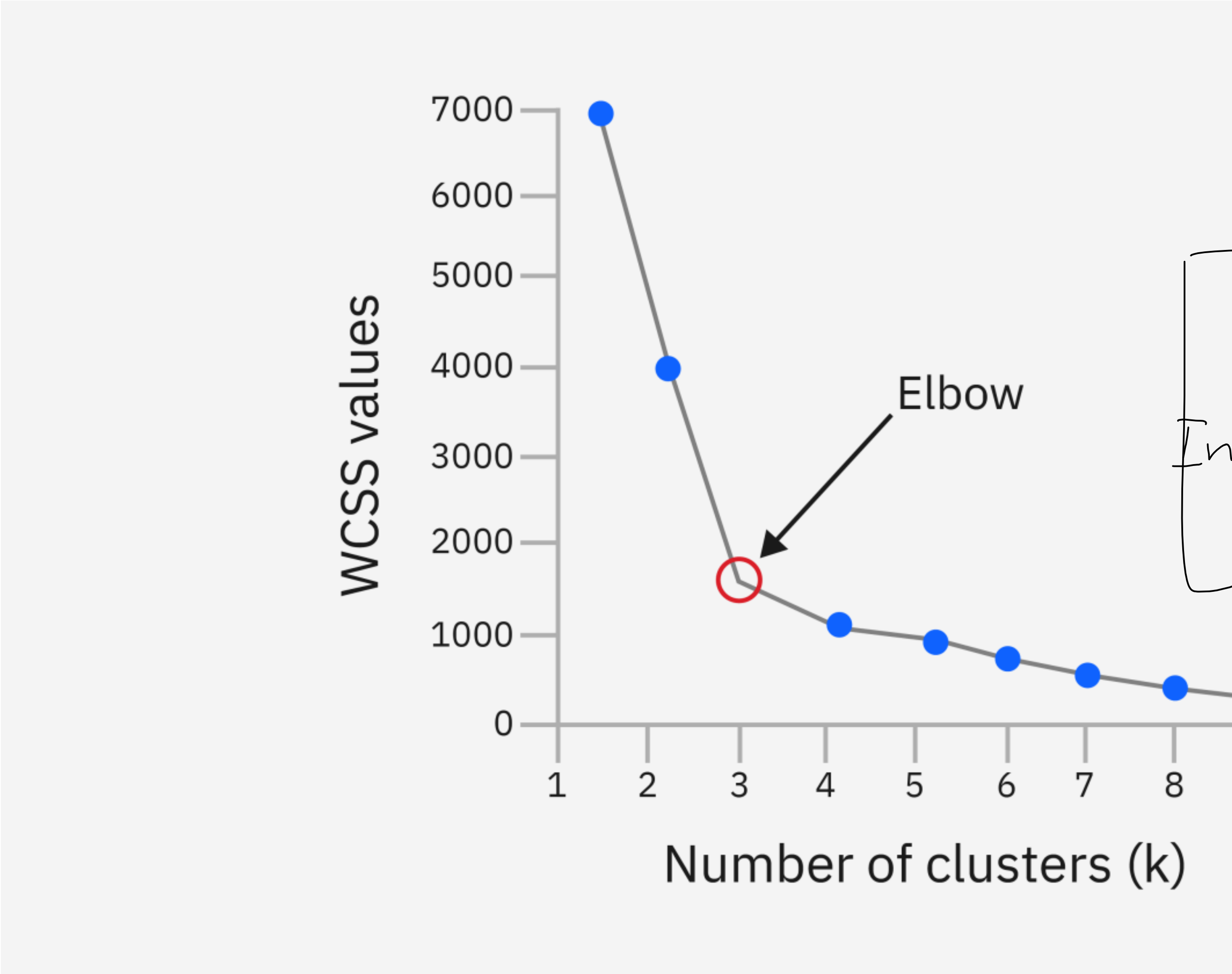
$K = 6 \quad K = 2$



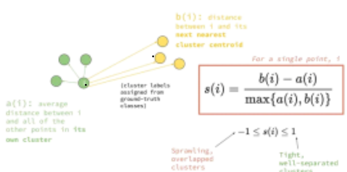
Compute the average dist to p a_i → a_i

compute average distance to p to C_i → b_i

$$\frac{b_i - a_i}{\max(a_i, b_i)} = 0 \rightarrow -1 \quad = [-1, 1]$$



7000
6000
5000
4000
3000
2000
1000
0
WCSS values
Elbow
1
2
3
4
5
6
7
8
Number of clusters (k)

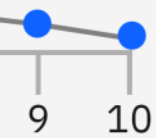


b(i): distance between i and its next nearest cluster centroid
For a single point, i
a(i): average distance between i and all of the other points in its own cluster
(cluster labels assigned from ground-truth classes)
$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$
$-1 \le s(i) \le 1$
Sprawling, overlapped clusters
Tight, well-separated clusters

+1
0
-1

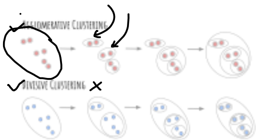
DBI

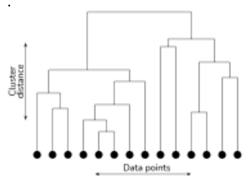
ertia = $\sum_{i=0}^{n} \min_{\mu_j \in C} \left(\|x_i - \mu_j\| \right)^2$

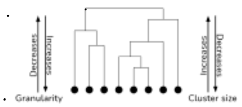


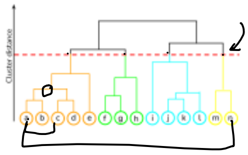
a

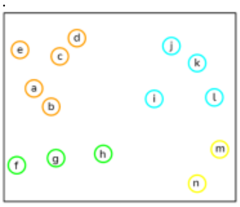
$$DBI = \frac{1}{N}\sum_{i=1}^{N} \max \frac{(S_i + S_j)}{d_{ij}}$$

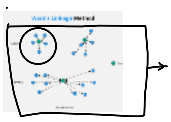


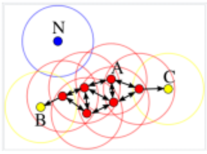




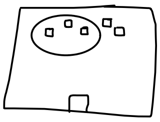








minpts
max distance



minsamp
EPS

Unsupervised

Clustering
[PCA
SVD
LDA] Stationary